CASE

62 year old man · sudden tearing chest pain radiating into the back · CTA confirms thoracic aortic dissection

BIO 004 · WEEK 1 · CASE DEEP DIVE

Body Cavities & Regions

The compartments organs live in, and the maps clinicians use to find what hurts

PATIENT CASE

A **62 year old man** arrives in the emergency department complaining of sudden onset severe chest pain that he describes as **tearing and radiating into his back**. His **blood pressure is elevated** and the emergency physician becomes concerned about an **acute aortic syndrome**, including an **aortic dissection**.

While the team stabilizes the patient and obtains IV access, imaging is ordered. Because he is hemodynamically stable, the diagnostic test of choice is **CT angiography of the chest with IV contrast**. The CT shows a **thoracic aortic dissection involving the descending aorta**. The radiologist also notes that the **mediastinum is widened** on the scout film.

> GUIDING QUESTION
>
> Which cavity does this pathology live in, which structures around it are at risk, and how would you describe the location of his pain to a colleague using only standardized anatomical terms?

PREWORK

This page is the case-based companion. The terms, the video, and the spaced recall for cavities, serous membranes, peritoneum, and the abdominal grids all live on your Week 1 prework page. Open both side by side: the prework gives you the inventory; this page gives you the story. The mediastinum gets its own full chapter linked from section 2.

1 WHERE THE PAIN LIVES

Organs don't float; they live in cavities

Our patient's pain is tearing and it moves to his back. Before we can say anything about what's wrong, we need to know **where** wrong is. The body is organized into a small set of named cavities, and every viscus you will ever care about sits inside one of them.

Two big groupings. The **dorsal cavity** protects the central nervous system: the cranial cavity holds the brain, the vertebral cavity holds the spinal cord, and three protective membranes called the meninges line them. The **ventral cavity** holds everything else, separated into the thoracic cavity above the diaphragm and the abdominopelvic cavity below.

Inside the thorax there are three subdivisions: a *right pleural cavity* for the right lung, a *left pleural cavity* for the left lung, and between them a central compartment called the *mediastinum* that carries the heart, the great vessels, the trachea, the esophagus, and the rest of the non-lung thoracic structures. Inside the mediastinum a smaller serous sac, the *pericardial cavity*, surrounds the heart.

Thoracic cavity

Above the diaphragm. Houses the two pleural cavities (lungs) and the mediastinum (everything that is not lung).

Diaphragm

Muscular floor of the thoracic cavity and roof of the abdominal cavity. The single most important boundary in the trunk.

Abdominal cavity

From the diaphragm to the pelvic brim. Most of the digestive tract, spleen, kidneys, adrenals.

Pelvic cavity

Below the pelvic brim. Bladder, reproductive organs, distal rectum.

Why this matters for our patient: the descending thoracic aorta sits in the mediastinum, against the spine. A dissection there spreads in a space with very little room and against neighbors that all matter.

2 THE COMPARTMENT UNDER TENSION

The mediastinum gets a chapter of its own

The mediastinum is the central thoracic compartment, flanked on each side by the pleural sacs. It carries the heart, great vessels, trachea, esophagus, thymus, thoracic duct, and a dense network of nerves and lymph nodes. Because so many vital structures share such a small space, a mass or a bleed here puts immediate pressure on neighbors.

Our patient's dissection lives in the **posterior mediastinum**, where the descending aorta sits against the vertebral bodies. That position explains the tearing pain that radiates into his back, and it explains why the radiologist saw a widened mediastinum on the scout film: blood inside the dissected aortic wall puffs out the central thoracic silhouette.

The mediastinum is its own world, and you'll want the full cross-sectional anatomy, the compartments, the imaging logic, and the differential-by-compartment for any mass you find here. That lives in its own deep-dive page: read the mediastinum chapter.

Serous membranes, the same design used three times

Three cavities in the body are lined by serous membranes: the **pleural** cavities around the lungs, the **pericardial** cavity around the heart, and the **peritoneal** cavity around most of the abdominal organs. The architecture is the same in all three. Learn it once and you understand all three.

Picture pressing your fist into a half-inflated balloon. The balloon wraps your fist in two layers. The inner layer, touching your fist, is the *visceral* layer. The outer layer, facing the room, is the *parietal* layer. The thin film of air between them is the cavity. Now swap the air for a small amount of clear lubricating *serous fluid*, and swap your fist for a lung, a heart, or a loop of intestine.

That is the entire trick. The lung slides against the inside of the chest wall every breath, the heart pumps against the inside of the pericardium every beat, and the intestines slither against the abdominal wall every minute of digestion, all without friction, because of a thin film of fluid trapped between two slippery membranes.

> **When the lubrication fails.** Inflammation of these membranes (pleuritis, pericarditis, peritonitis) makes the two layers grate against each other, which is sharply painful. Clinicians can sometimes *hear* the grating with a stethoscope as a friction rub.

CAVITY	PARIETAL LAYER	VISCERAL LAYER	INFLAMMATION
Pleural	lines the thoracic wall and superior diaphragm	covers the lung surface (visceral pleura)	pleuritis (pleurisy)
Pericardial	parietal pericardium, fused to the fibrous pericardium	visceral pericardium (epicardium) on the heart surface	pericarditis
Peritoneal	lines the abdominopelvic wall	covers the abdominal organs	peritonitis

DRAW ZONE 1

✎ -- *Sketch a midsagittal body. Label dorsal and ventral cavities. Then draw the fist-in-balloon for a serous membrane.*

Mark the diaphragm clearly; it is the most clinically important boundary.

The "inside vs behind" distinction that changes everything

The peritoneum wraps most of the abdominal organs the same way the pleura wraps the lungs. But not every organ is inside the peritoneal sac. Some sit behind it, against the back wall of the abdomen, covered by peritoneum only on their anterior surface. That distinction is called **intraperitoneal** vs **retroperitoneal**, and it changes the surgery, the imaging, and the way the patient presents.

Why it matters for our case: the descending aorta is **retroperitoneal**. When it dissects or bleeds in the abdomen, blood collects behind the parietal peritoneum, not inside it. The patient's belly may stay soft and non-tender even while they hemorrhage. A perforated stomach (intraperitoneal) is the opposite story: gastric contents spill into the peritoneal cavity and the abdomen rapidly becomes rigid and exquisitely painful.

The mnemonic worth memorizing for the major retroperitoneal organs is **SAD PUCKER**: Suprarenal (adrenal) glands, Aorta and IVC, Duodenum (most), Pancreas (except tail), Ureters, Colon (ascending and descending), Kidneys, Esophagus (abdominal portion), Rectum (distal third).

Inside the peritoneal cavity, the membrane forms folds and aprons with names you'll meet again: the *mesentery* suspends the small intestine from the back wall and carries its blood supply; the *greater omentum* is the fatty apron hanging off the stomach that migrates toward infection and helps wall it off; the *falciform ligament* tethers the liver to the anterior abdominal wall.

5 MAPS OF THE ABDOMEN

Two grids: one for precision, one for speed

The abdomen sits under one continuous skin surface, but two named grids let clinicians point at exactly the same place every time. The **nine-region grid** is the anatomist's map: two midclavicular lines and two horizontal planes (subcostal and intertubercular) divide the abdomen into nine named regions, each one sitting over specific organs.

The **four-quadrant grid** is the bedside language: two lines, one vertical through the midline and one horizontal through the umbilicus, divide the abdomen into RUQ, LUQ, RLQ, and LLQ. It is faster, less precise, and overwhelmingly the system clinicians actually document in the chart. "Right lower quadrant pain with rebound" is appendicitis until proven otherwise; "right iliac region tenderness" says the same thing in the anatomy textbook.

RUQ — right upper quadrant

Liver, gallbladder, right kidney, hepatic flexure, head of pancreas, duodenum. Think biliary colic, cholecystitis, hepatitis.

LUQ — left upper quadrant

Spleen, stomach, left kidney, splenic flexure, tail of pancreas. Think splenic injury, pancreatitis, gastric pathology.

RLQ — right lower quadrant

Cecum, appendix, right ovary, right ureter. Think appendicitis, ectopic pregnancy, ovarian cyst, ureteral colic.

LLQ — left lower quadrant

Sigmoid colon, descending colon, left ovary, left ureter. Think diverticulitis, ovarian pathology, ureteral colic.

✎ **DRAW ZONE 2** -- *Sketch an anterior abdomen. Overlay the nine-region grid in one color and the four-quadrant cross in another. Mark where a dissecting aorta would sit on a transverse cut.*

Use Fine for the dissection flap inside the aortic wall.

RETURNING TO THE TRAUMA BAY

Walk the dissection: cavity, membrane, neighbors

You started with a 62-year-old man with tearing pain that radiated to his back. Now you can name exactly where the pathology lives and what sits around it.

"Where does the dissection live?"

Thoracic cavity, posterior mediastinum. The descending aorta is a posterior mediastinal structure that sits against the spine.

"What membranes are at risk?"

Parietal pleura on both sides flank the aorta. The pericardium is at risk if the dissection extends superiorly.

"Which neighbors could be compressed?"

Esophagus anteriorly, thoracic duct posteriorly, vertebral bodies behind, azygos vein to the right, sympathetic trunks paravertebral.

"Why does the pain go to the back?"

The posterior mediastinum sits directly against the spine. Expansion or hemorrhage there refers pain posteriorly.

"What if the dissection extends downward?"

Into the abdominal aorta, which is retroperitoneal. Blood collects behind the parietal peritoneum, so the abdomen may stay soft even as the patient hemorrhages.

"Why a widened mediastinum on imaging?"

Blood inside the dissected aortic wall increases the central thoracic silhouette on chest X-ray. CT angiography then shows the dissection flap directly.

Cavity, membrane, neighbors. Every patient who arrives in the emergency department can be described that way. When you go back to the prework page, the cavities and the peritoneal relationships will land differently now, because each one points at a real space where something real can go wrong.